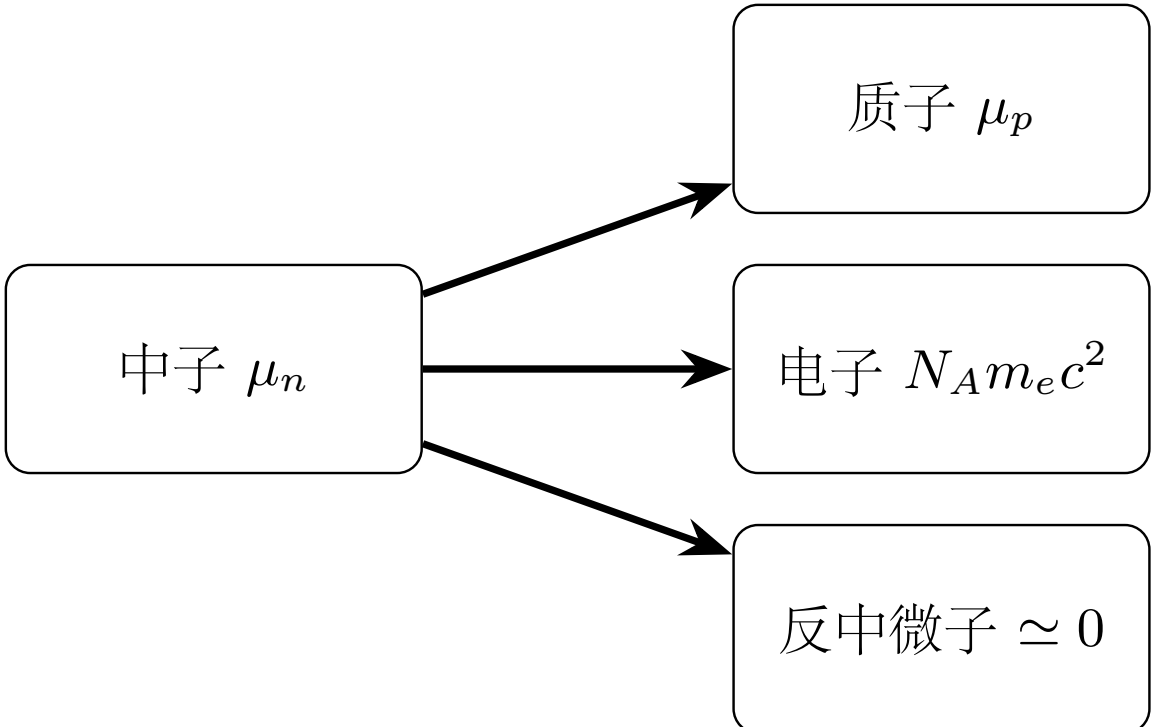


$$RT \ln \frac{1 - x_p}{x_p} = N_A m_e c^2$$

$$x_p = \frac{1}{1 + e^{N_A m_e c^2/(RT)}}$$